

SANA: LakeQA SUBSET20B results

Experiment records 2026-08-31-web-arm-subset20b and 2026-08-31-model-tiers-subset20b

1 Results

1.1 Retrieval source: open web versus the data lake

To test whether the retrieval axis is specific to the data lake or reflects a more general property of the agent, we replaced lake search with open-web search. The arm uses the Parallel Search API and, under `--no-s3`, loses every lake-backed file tool: the agent retains only `download`, which fetches an arbitrary `http(s)` URL into its sandbox, and `execute_code`. All four arms hold the remaining axes fixed at `search_results=naive`, `profile=standard` and `computation_tool=standard`, since the ideal computation tools resolve against authored data-lake records and would make a web run’s outcome independent of what it retrieved.

Table 1 reports the result. The web arm reaches 35.0% semantic match against 45–50% for the three lake arms, and the three lake conditions are separated by a single task. The interesting quantity is not the gap in accuracy but the gap in retrieval: the ideal arm reaches 78.1% of the gold sources using 22.2 cycles, while the standard arm reaches 67.9% using 29.2 cycles and 56% more money. Oracle retrieval is markedly more efficient and yet converts to the same number of correct answers, which locates the bottleneck downstream of retrieval on this split.

Two properties of the web arm complicate its interpretation and we report them rather than correct for them. First, `download` accepts any URL, and the model routinely constructs one from pretrained knowledge rather than following a search result — live Socrata queries against `data.cityofnewyork.us`, `data.va.gov` export endpoints, and in one case a timestamped `web.archive.org` URL. The arm therefore measures an agent with open web access and code execution, not web-search retrieval, and its data path may be *stronger* than the lake arms’, since it aggregates live authoritative sources rather than a fixed snapshot. Second, that liveness produces false negatives: two failures are arithmetically correct against current data but disagree with gold answers derived from the snapshot (2,700,968 against a gold 2,705,988; 2923.59 against 2941).

A third of all task-runs (33 of 80) reached the 30-call tool budget, and every arm scored roughly two to three times better below the cap than at it, so the absolute numbers here should be read as depressed by the budget rather than as ceilings.

Table 1: Retrieval-source comparison on LakeQA SUBSET20B (gpt-5-mini, 20 tasks per arm, zero errored runs). SM is semantic match (judge gpt-5.4); EM is lexical exact match. *Blank* counts answers the judge classified as `answer_unknown_blank`, i.e. honest abstentions rather than wrong answers. *Src* is the fraction of required gold sources the agent read; the web arm touches no lake sources by construction. *Cap* counts tasks that reached the 30-call tool budget.

Arm	SM (%)	EM (%)	Blank	Cycles	Tools	Cap	Cost (\$)	Src (%)
Ideal	50.0	45.0	0	22.2	21.8	5	0.60	78.1
Standard	50.0	45.0	0	29.2	28.2	11	0.94	67.9
Naive	45.0	45.0	0	27.8	26.8	9	0.83	70.1
Web	35.0	30.0	1	26.2	25.2	8	1.00	—

Semantic match versus exact match. The two metrics differ by exactly one task, and that task is instructive. For the question whose gold answer is “*the Cut Bank Penguin*”, three arms answered [Cut Bank

Penguin] and the fourth answered [World's largest statue...]. All four are correct; all four score zero under exact match, because the lexical metric penalises a dropped article. On LakeQA most gold answers are a single token, where token-overlap F1 is identical to exact match by construction, and where it does differ it largely measures whether the model reproduced determiners. We therefore report semantic match throughout and do not report F1.

1.2 Replicate variance on the ablation grid

We ran the leave-one-out ablation over three model tiers on SUBSET20B, a 20-task split of LakeQA stratified jointly on node count and historical solve rate. Every axis is held at the oracle and one axis is degraded per cell, giving seven cells per model. Each cell was run **three times** under identical configuration, so every number carries a standard deviation rather than resting on a single sample.

That replication is the central result. The mean per-cell standard deviation is 6.5 pp, so a two-sigma interval spans ± 13 pp and an effect must exceed roughly 13 pp to be distinguishable from run-to-run variance. Since one task is worth 5 pp at $n = 20$, that threshold is about 3 tasks.

Of the 18 degraded cells, only 7 clear that threshold: Search: BM25 at **gpt-5.4-nano** (-23.3 pp), Plan: Default at **gpt-5.2** (-21.7 pp), Search: PNEUMA at **gpt-5-mini** (-16.7 pp), Search: BM25 at **gpt-5.2** (-16.7 pp), Search: BM25 at **gpt-5-mini** (-15.0 pp). The search axis is the one effect that replicates across every tier; the remaining cells cannot be separated from noise at this sample size.

The reference cells themselves are **gpt-5.4-nano** 36.7% ($\sigma = 2.9$), **gpt-5-mini** 68.3% ($\sigma = 2.9$), **gpt-5.2** 68.3% ($\sigma = 7.6$). **gpt-5-mini** and **gpt-5.2** land on the same mean (68.3%), despite a roughly sevenfold difference in price per token. On this split the frontier model does not outperform the smaller one, and a single round would have suggested otherwise: round 1 alone put **gpt-5.2** 5 pp below **gpt-5-mini**.

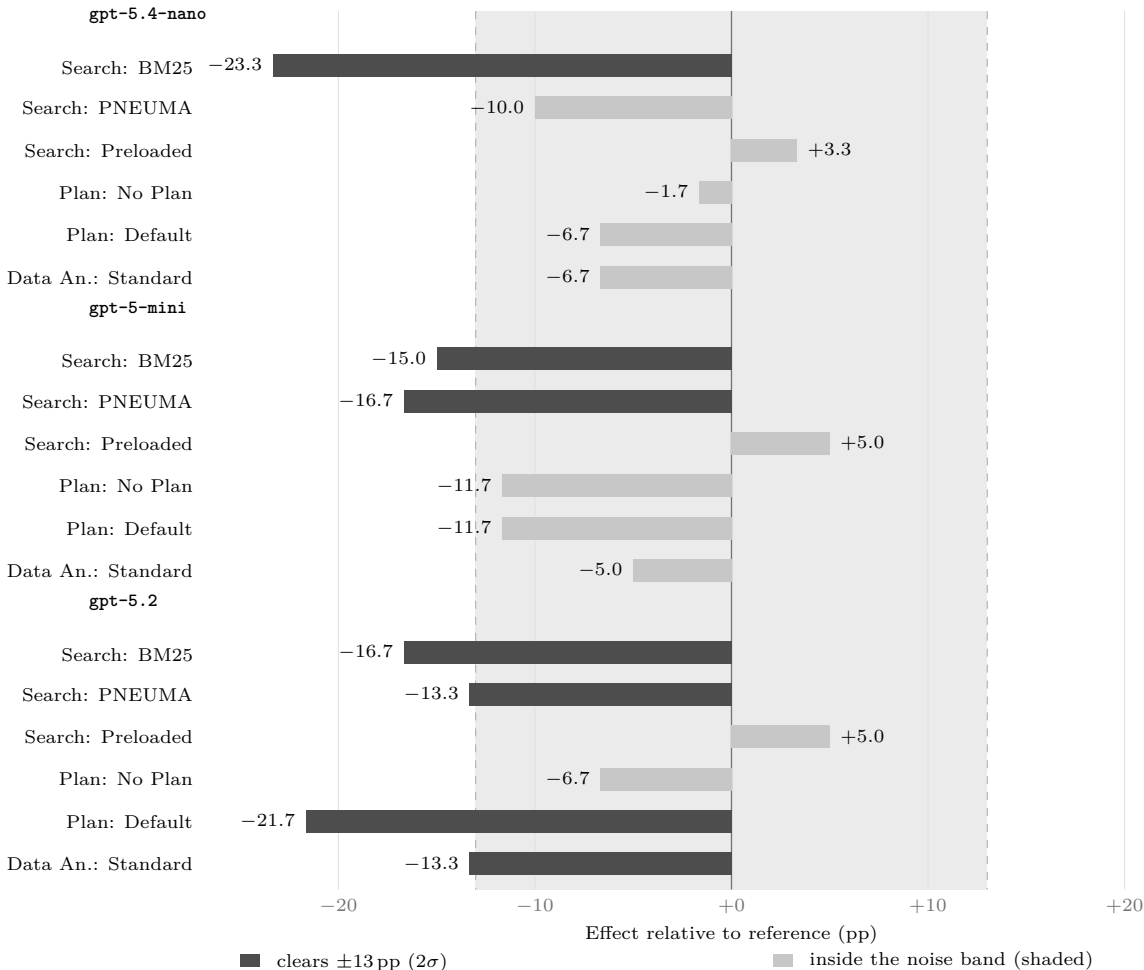


Figure 1: Ablation effects across three replicate rounds. Each bar is the change from that model’s all-ideal reference when one axis is degraded; the shaded column is the ± 13 pp two-sigma noise band. Bars ending inside it cannot be distinguished from run-to-run variance at $n = 20$, where one task is worth 5 pp.

Table 2: Leave-one-out ablation across model tiers on LakeQA SUBSET20B (20 tasks per cell, exact match, three replicate rounds under identical configuration). $\bar{x}$ is the mean over rounds and σ their standard deviation; δ is the effect relative to that model’s reference cell. Effects exceeding the ± 13 pp two-sigma threshold are set in bold.

Model	Condition	R1	R2	R3	$\bar{x}$ (%)	σ	δ (pp)
gpt-5.4-nano	Reference (all ideal)	35	40	35	36.7	2.9	—
	Search: BM25	20	15	5	13.3	7.6	−23.3
	Search: PNEUMA	30	30	20	26.7	5.8	−10.0
	Search: Preloaded	45	40	35	40.0	5.0	+3.3
	Plan: No Plan	25	45	35	35.0	10.0	−1.7
	Plan: Default	40	35	15	30.0	13.2	−6.7
	Data An.: Standard	35	25	30	30.0	5.0	−6.7
gpt-5-mini	Reference (all ideal)	65	70	70	68.3	2.9	—
	Search: BM25	50	65	45	53.3	10.4	−15.0
	Search: PNEUMA	45	50	60	51.7	7.6	−16.7
	Search: Preloaded	80	70	70	73.3	5.8	+5.0
	Plan: No Plan	60	55	55	56.7	2.9	−11.7
	Plan: Default	55	55	60	56.7	2.9	−11.7
	Data An.: Standard	60	70	60	63.3	5.8	−5.0
gpt-5.2	Reference (all ideal)	60	75	70	68.3	7.6	—
	Search: BM25	55	50	50	51.7	2.9	−16.7
	Search: PNEUMA	60	60	45	55.0	8.7	−13.3
	Search: Preloaded	80	70	70	73.3	5.8	+5.0
	Plan: No Plan	70	60	55	61.7	7.6	−6.7
	Plan: Default	55	40	45	46.7	7.6	−21.7
	Data An.: Standard	65	50	50	55.0	8.7	−13.3

Table 3: Replicate noise floor, over every cell run three times. One task is worth 5 pp at $n = 20$, so the mean per-cell standard deviation is close to a single task and the two-sigma threshold is about 3.

Statistic over per-cell σ	Value (pp)
Mean	6.5
Median	5.8
Maximum	13.2
Two-sigma threshold	13.0
Cells	21